

Titration according to Kjeldahl: Test with Ammonium Sulfate Solution

Sample: Water solution of ammonium sulfate 0.06 ... 0.07 g/50 mL

Substance: Ammonium sulfate $(\text{NH}_4)_2\text{SO}_4$
M = 132.14 ; z = 2

Preparation: – max. 0.07 g in 50 mL flask;
– 20 mL deion. water;
– 15 mL sulfuric acid, H_2SO_4 (98%).

Titrant: Hydrochloric acid, HCl
c(HCl) = 0.1 mol/L

Standard: Tris(hydroxymethyl)amino-
methane (THAM),
0.05 ... 0.1 g / 50 mL H_2O .

Instruments: METTLER DL67 DL70ES DL77
METTLER ST20A Changer
METTLER TOLEDO AT261
Printer HP DeskJet 510
Water steam distilling unit

Accessories: Titration beaker ME-101974
Peristaltic Pump ME-65241
1 Burette drive DV90
Temp. sensor DT120 ME-89624

Indication: DG111-SC at Sensor 1

Results:

METTLER DL77		Titration	V3.1	Mettler-Toledo AG	
		DL77/115		Application Laboratory	
Method	KKA2 Kjeldahl / Titration			22-Mar-1995	15:17
User	cdc				
Measured	22-Mar-1995 17:54				
RESULTS					
No	ID1	ID2	Sample amount and results		
1/1	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3269	mmol/L	(NH ₄) ₃ BO ₃
			R2 = 0.6870	mg	N
			R3 = 13.7408	mg	N Total
1/2	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3270	mmol/L	(NH ₄) ₃ BO ₃
			R2 = 0.6872	mg	N
			R3 = 13.7447	mg	N Total
1/3	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3270	mmol/L	(NH ₄) ₃ BO ₃
			R2 = 0.6873	mg	N
			R3 = 13.7457	mg	N Total
1/4	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3262	mmol/L	(NH ₄) ₃ BO ₃
			R2 = 0.6855	mg	N
			R3 = 13.7108	mg	N Total

1/5	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3266	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
			R2 = 0.6864	mg	N
			R3 = 13.7289	mg	N Total
1/6	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3273	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
			R2 = 0.6878	mg	N
			R3 = 13.7561	mg	N Total
1/7	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3260	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
			R2 = 0.6851	mg	N
			R3 = 13.7027	mg	N Total
1/8	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3266	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
			R2 = 0.6863	mg	N
			R3 = 13.7261	mg	N Total
1/9	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3264	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
			R2 = 0.6860	mg	N
			R3 = 13.7197	mg	N Total
1/10	Amm.Bor.	Test 10	50.0 mL	g	Fixed volume U
			R1 = 0.3273	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
			R2 = 0.6877	mg	N
			R3 = 13.7548	mg	N Total
STATISTICS					
Number results		R1	n = 10		
Mean value			$\bar{x}$ = 0.3267	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
Standard deviation			s = 0.000436	mmol/L	$(\text{NH}_4)_3\text{BO}_3$
Rel. standard deviation		srel = 0.133	%		
Outlier test: no outliers!					
STATISTICS					
Number results		R2	n = 10		
Mean value			$\bar{x}$ = 0.6867	mg	N
Standard deviation			s = 0.000915	mg	N
Rel. standard deviation		srel = 0.133	%		
Outlier test: no outliers!					
STATISTICS					
Number results		R3	n = 10		
Mean value			$\bar{x}$ = 13.7330	mg	N Total
Standard deviation			s = 0.018309	mg	N Total
Rel. standard deviation		srel = 0.133	%		
Outlier test: no outliers!					

Method

Method	KKA2	Kjeldahl / Titration
Version	22-Mar-1995	15:17
Title		
Method ID	KKA2	
Title	Kjeldahl / Titration	
Date/time	22-Mar-1995 15:17	
Sample		
Number samples	10	
Titration stand	ST20A 1	
Entry type	Fixed volume U	
Volume [mL]	50.0	
ID1	Amm.Bor.	
Molar mass M	112.96	
Equivalent number z	3	
Temperature sensor	TEMP A	
Titration		
Titrant	HCl	
Concentration [mol/L]	0.1	
Sensor	DG111-SC	
Unit of meas.	As installed	
Titration mode	EQP	
Titrant addition	DYN	
ΔE(set) [mV]	8.0	
Limits ΔV	Absolute	
ΔV(min) [mL]	0.02	
ΔV(max) [mL]	0.3	
Measure mode	EQU	
ΔE [mV]	1.0	
Δt [s]	2.0	
t(min) [s]	5.0	
t(max) [s]	30.0	
Threshold	100.0	
Maximum volume [mL]	30.0	
Termination after n EQPs	Yes	
n =	1	
Evaluation procedure	Standard	
Calculation		
Result name	(NH ₄) ₃ BO ₃	
Formula	R=Q*C/U	
Constant	C=1000/z	
Result unit	mmol/z	
Decimal places	4	
Calculation		
Result name	N	
Formula	R2=R1*(U/1000)*C2	
Constant	C2=z*14.01	
Result unit	mg	
Decimal places	4	
Calculation		
Result name	N Total	
Formula	R3=R2*20	
Constant		
Result unit	mg	
Decimal places	4	
Record		
Output unit	Printer	
Raw results last sample	Yes	
Table of values	Yes	
E - V curve	Yes	
Rinse		
Auxiliary reagent	H ₂ O	
Volume [mL]	20.0	
Statistics		
Ri (i=index)	R1	
Standard deviation s	Yes	
Rel. standard deviation srel	Yes	
Outlier test	Yes	
Statistics		
Ri (i=index)	R2	
Standard deviation s	Yes	
Rel. standard deviation srel	Yes	
Outlier test	Yes	
Statistics		
Ri (i=index)	R3	
Standard deviation s	Yes	
Rel. standard deviation srel	Yes	
Outlier test	Yes	
Record		
Output unit	Printer	
All results	Yes	

Author: C. De Caro

Sample preparation

1) Ammonium sulfate stock solution:

- Prepare a stock solution of ca. 0.7 g ammonium sulfate in a 500 mL flask and fill with 200 mL deionized water;
- Place the 500 mL flask in a cold water bath and add **slowly** 150 mL 98% sulfuric acid (Merck, for nitrogen determination, Nitrogen max. 0.0005%); (**CAUTION: hot !**)
- Allow the solution to cool down to room temperature and fill to the mark with deion. water.

2) Distillation of ammonia:

To distill ammonia it is necessary to use an *appropriate distilling unit* with a water steam generator, e.g. Kjeltec 1002 -Tecator- or Büchi 316 Distilling Unit.

- Place 50 mL 1.5% boric acid solution (H₃BO₃) in a flask at the **outlet** of the distilling unit and add 3 drops of mixed indicator 5 (Merck). Be sure that the delivery tube of the distilling unit is **dipped into** the boric acid solution;
- Pipette 50 mL of the ammonium sulfate solution into the appropriate glass tube for distillation;
- Add ca. 50 mL water and fix the glass tube to the **inlet** of the distilling unit;
- Add ca. 60-80 mL 30% sodium hydroxide solution (NaOH) and **immediately** start the distillation by opening the valve of the water steam generator;
- When the indicator colour has changed from *violet to green*, wait ca. 1 minute and then stop the distillation.

Literature: see ref. 1, 2, 3.

3) Titration:

- Pour the distilled sample into a 250 mL flask (or 500 mL or 1000 mL) and fill up to the mark with deionized water;
- Pipette 50 mL into a titration beaker and start the method KKA2 (automated titration with 0.1 mol/L HCl).

Comments to Method

- 1) To check whether the 30% sodium hydroxide solution added is **enough** for the ammonia distillation, it is very helpful to put 1-3 drops of mixed indicator 5 in the sample (colour change from violet to green);
- 2) The neutralization of sulfuric acid with concentrated alkaline solution leads to a strong heating of the sample. To avoid this, deionized water is added before starting the distillation step (see sample preparation, 2c).
- 3) The distilled sample was diluted with deionized water in a 1000 mL flask. Thus, the corresponding factor (20) was taken into account in the calculation R3.

General Remarks

- 1) A blank determination did not give any results.
- 2) The analysis time can be decreased by predispensing (see function "TITRATION")

Waste disposal

After titration, the weakly acidic samples should be neutralized before final disposal.

Titration Curve

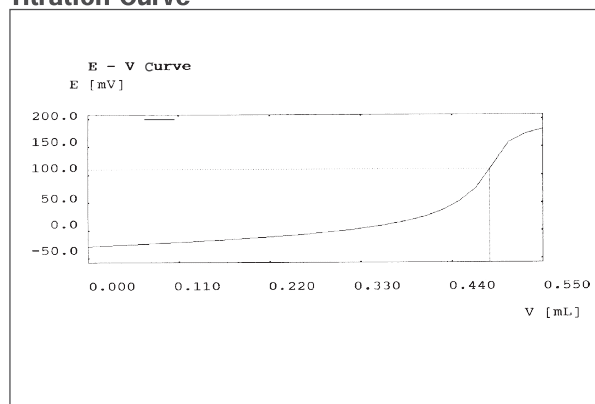


Table of Measured Values

	Volume mL	Increment mL	Signal mV	Change mV	1st deriv. mV/mL	Time min:s
ET1	0.0000		-29.3			0:11
	0.0200	0.0200	-27.7	1.6	80.5	0:16
	0.0400	0.0200	-26.2	1.5	73.5	0:21
	0.0800	0.0400	-23.7	2.5	63.0	0:26
	0.1600	0.0800	-17.7	6.0	75.2	0:34
	0.2510	0.0910	-8.6	9.1	100.0	0:43
	0.3150	0.0640	0.4	9.0	141.1	0:52
	0.3580	0.0430	9.0	8.5	198.6	1:01
	0.3890	0.0310	18.0	9.0	291.3	1:10
	0.4090	0.0200	25.3	7.3	364.0	1:19
	0.4290	0.0200	36.6	11.3	567.0	1:29
	0.4490	0.0200	52.4	15.8	791.0	1:41
	0.4690	0.0200	75.2	22.8	1141.0	1:50
	0.4890	0.0200	116.1	40.8	2040.5	2:03
EQP1	0.5090	0.0200	158.1	42.1	2103.5	2:17
	0.5290	0.0200	173.7	15.5	777.0	2:29
	0.5490	0.0200	181.6	7.9	395.5	2:38

Results of Series (Titration)

Range of Sample Concentration (mg N / 50 mL)	n	Recovery %	Reproducibility %
2 – 3	4	98.90	0.036
	4	99.39	0.080
	4	99.39	0.106
	4	99.31	0.123
1 – 2	5	99.11	0.113
	4	98.92	0.085
0.5 – 1	10	98.60	0.226
	8	98.53	0.075
	10	98.29	0.133
	9	97.99	0.136
	9	99.02	0.220